

Marlon Buchanan

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Industry Experience

R&D Engineering Co-op, FTXT Energy Technology – North Vancouver, BC Sept 2023 – Apr 2024

- Characterized fuel cell plate resistance under compression using a UTM, recording voltage drop per layer for thermal resistance network modeling
- Automated life-cycle mechanical analysis in MATLAB, generating 120+ FEA load profiles for GB/T 28046.3 compliance, reducing analysis time by 50%
- Built a Python scraper to extract conductivity, tensile strength, yield stress, and elastic modulus from manufacturer datasheets into a materials database (75+ metals and plastics) for FEA simulations
- Verified GD&T compliance on 5+ parts against technical drawings, catching a dimensional fault before shock and vibration testing

Engineering Projects

Self-Balancing Robot (Edmond) - STM32, C, MATLAB/Simulink, SolidWorks Jun 2026 – Present

- Built and 3D-printed a SolidWorks chassis integrating motors, electronics, and OLED; robot balances unaided and rejects disturbances up to $\pm 10^\circ$
- Derived a discrete LQR controller in MATLAB/Simulink from a linearized cart-pendulum model, with physical parameters from SolidWorks mass properties
- Implemented complementary filter sensor fusion on STM32F303K8 in HAL C, fusing MPU6050 accel/gyro data over I²C for real-time pitch estimation
- Diagnosed intermittent I²C failures from loose connectors; wrote a modular driver with bus-recovery logic and soldered the final build to eliminate the fault

Solar MPPT Converter PCB - Altium, LTspice, PLECS, MSP430 Sept 2025 – Apr 2026

- Designed a synchronous buck converter for solar MPPT: 0–42V input, 3.7V output, 500kHz switching, sized for low-current organic solar panels
- Diagnosed a bring-up failure on the initial buck-boost topology and redesigned to a synchronous buck with half-bridge driver, simplifying gate drive and cutting assembly risk
- Laid out a 4-layer PCB for staged bring-up with scope test points, ground references, and 0-ohm bridges; iterated across 3 board spins within a 4-month schedule
- Hand-assembled 8+ breakout boards across power stage, gate drivers, I/V sensing, and battery charging; verified with DMM and oscilloscope

Education

Simon Fraser University

Bachelor of Applied Science, Sustainable Energy Engineering

Expected Sept 2027

Relevant Courses: *Embedded Computer Systems, Electronic Devices and Systems, Feedback Control Systems, Power Electronics, Advanced Power Electronics*

Technical Skills & Interests

Design Tools: Altium Designer, SolidWorks, LTspice, PLECS, MATLAB/Simulink

Embedded/Hardware: STM32 (Cortex-M4), Arduino (ATmega328P), MSP430, I²C, UART, SPI, PWM

Programming: C, C++, Python

Lab Tools: Oscilloscope, DMM, Function Generator, Power Supply, Soldering Iron, 3D-Printing, Wire Crimp

Interests: Street Photography, Psychology Books, Running, Hot Yoga, Bodybuilding, Hiking, Volleyball